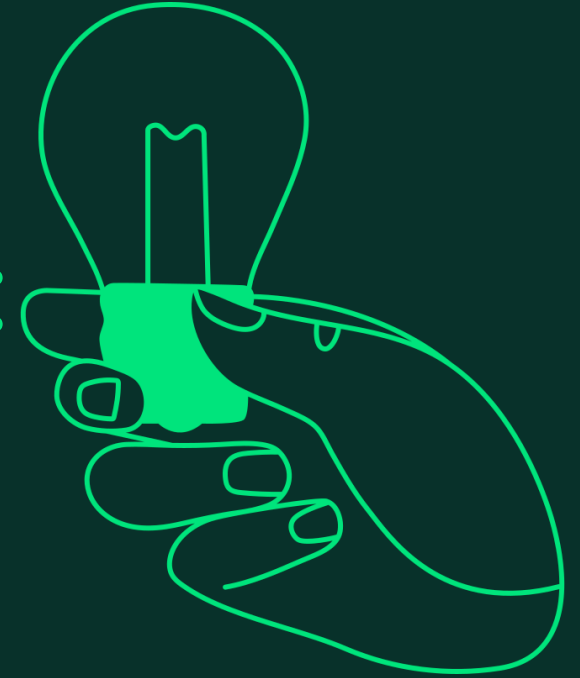


Interplay Between Statistical Design and Operational Execution: Points to consider



Dooti Roy, Head of Innovation, Clinical Data Science and Methodology
On Behalf of
Efficiency+ Sub-team of Design and Operation

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Efficiency is becoming the common currency of drug development

The mandate has shifted from “faster” to “faster with rigor, relevance and access.”

01 REGULATORS

Remove unnecessary burden while preserving participant protection and decision-quality evidence.

02 SPONSORS

Improve productivity through **smarter design**, simpler processes, better data flow, and shorter cycle times.

03 PATIENTS

Reduce participation burden and avoid inconclusive trials, enabling earlier and more affordable access.

RECENT SOURCE SIGNALS

U.S. FDA • Operation Trailblazer 2026

“Eliminating unnecessary regulatory burden” while protecting participants and increasing submission quality.

EMA / ACT EU • November 2025

A shared ambition to make clinical research “better, faster and smarter” across the EU.

IQVIA Institute • Global Trends in R&D 2025

Efficiency is often seen as a integrated portfolio metric e.g., probability of success, trial intervals, enrollment, and cycle time, rather than simply reducing operational cost.

SHARED OUTCOME Less avoidable complexity. More useful evidence. Faster patient impact.

Biostatisticians can redesign the development pathway

Smarter Designs put decision points inside the design, then quantify the operational consequences.

SEAMLESS PHASE II/III

Select dose or treatment and transition without restarting the evidence stream.

IMPACT: 16.6–27.3% fewer participants in simulated oncology dose-optimization scenarios.¹

PLATFORM / MAMS

Share controls, remove unpromising arms and introduce new comparisons.

IMPACT: Platform comparisons achieved equal or better precision than separate trials, even with fewer participants.³

ADAPTIVE ENRICHMENT

Learn which biomarker-defined population benefits and focus recruitment.

IMPACT: Frequentist testing can preserve study-wise Type I error while Bayesian models guide enrollment decisions.⁵

SEQUENTIAL / RISK–BENEFIT

Stop, continue or select using joined efficacy and safety evidence.

IMPACT: MAMS designs can provide more power for the same sample size than alternative designs.⁷ Risk–benefit MAMS selection strongly controlled FWER with sample sizes like efficacy-only designs.⁸

FOCUS TO DATE These innovations have focused primarily on sample size, Type I error control and statistical power.

The methods are mature. Adoption is still selective.

Registry-inclusive reviews show pockets of uptake, concentrated by phase, disease area and crisis context, rather than routine use across development.

SEAMLESS PHASE II/III

27 of 317 adaptive-trial publications

8.5% of the searched literature

87% of adaptive trials were Phase I or II

Published studies searched: 2010–Apr 2020

PLATFORM (MAMS) TRIALS

98 unique randomized platform trials
identified **between 2015 - 2022**

68.3% registered in 2020–2022

Searched ClinicalTrials.gov, MEDLINE, EMBASE and
CENTRAL; uptake surged during COVID-19

Regulatory Opinion

FDA and EMEA consider such designs acceptable in the confirmatory stage as long as they can deliver appropriate-quality evidence with adequate control of bias, error, and interpretability

THE ADOPTION PARADOX Decades of methods → dozens of seamless studies → roughly one hundred platform programs

OUR HYPOTHESIS: The bottleneck has shifted from statistical invention to **operational readiness**

What does operational readiness look like?

The critical questions between a design that works on paper and a trial that works in practice

Where does information need to flow, and where does it materially change execution?

Which design features are easy to implement, and which create disproportionate cost?

Does adaptive decision-making create drug waste, resupply risk, or IRT (Interactive Response Technology) complexity?

Can site staff understand and execute the design reliably?

How much adaptiveness can operations absorb before execution becomes infeasible?

Is site initiation under more complex designs become more costly than recruiting additional patients?

Does the sample-size saving outweigh start-up delay, amendments and execution friction?

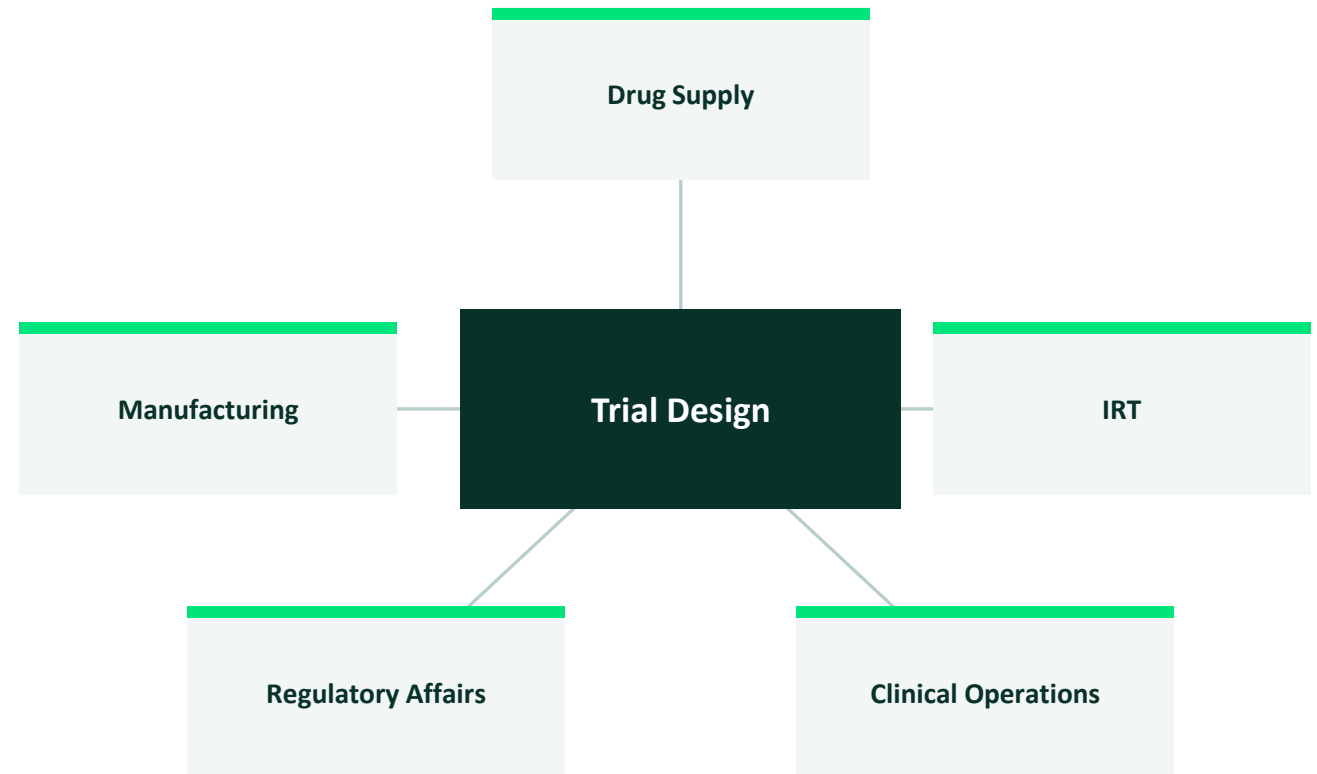
What must be known at design time, and what can remain adaptable?

OPERATIONAL READINESS A design whose information flow, cost, supply, sites, and decisions can move together.

Statistical trial design through the lens of feasibility and value

Statisticians often design innovative (adaptive) trials without fully understanding the impact of such designs on other functions such as clinical operations, manufacturing, regulatory affairs, drug supply, or IRT.

Trial design sits inside an operating system of functions that must move together.



MUTUAL UNDERSTANDING

Deepening mutual understanding can help improve internal acceptance, adoption, and ease implementation of such designs.

From awareness to implementation insight

A staged learning plan for seamless Phase II/III and platform trial execution

01

BUILD A COMMON LANGUAGE

Educate & create awareness

Cross-functional learning for statistics, clinical operations, supply, data, technology, regulatory and sites.

Shared vocabulary

WHAT COULD THESE DESIGNS ENABLE?

02

MAP THE LANDSCAPE

Broad organizational survey

Open to willing organizations. Measure familiarity, perceived relevance and actual implementation experience.

Experience-led sample

HOW WIDESPREAD IS READINESS?

03

UNDERSTAND THE HOW

Focused expert deep-dive

Engage colleagues with practical experience to map information flow, complexity, constraints, challenges and opportunities.

Information-flow blueprint

WHAT MAKES EXECUTION WORK?

OUTPUT

Operational readiness map

- Easy vs costly adaptations
- Supply, site & system impacts
- Execution risks & enablers
- Priority evidence gaps

ACTIONABLE AGENDA

EVIDENCE-GENERATION PLAN Learn broadly → identify experience → investigate deeply → translate into guidance

Considerations of a seamless 2-3 design on Clinical Operations (example)

When a Clinical Operations colleague worked through a seamless Phase 2-3 design:

SEVEN BUCKETS OF CONSIDERATIONS

01 Protocol and Design Readiness (Operational and Asset Lens)

02 Governance, Documentation, Decision-making, Transition Ownership and Inspection Readiness

03 Country, Site, and Patient Strategy (Continuity Focus)

04 Recruitment, Scalability, and Differentiation

05 Data Flow, Interim Analyses, and Systems Readiness

06 Budgeting, Financial Governance, and Burnout Risk

07 Communication and Change Management



BUCKET 01, UNPACKED

Protocol and Design Readiness

- Clearly defined Phase 2 to Phase 3 transition criteria
- Operational feasibility of amendments without enrollment halts, or minimized pauses in enrollment
- Alignment between statistical assumptions and real-world site, patient, and system behavior
- Forward planning for cumulative amendment burden (training, contracts, systems, TMF)
- Early clarity on what must be learned now vs what can safely wait to reduce asset risk
- Designing operational flexibility upfront to preserve strategic optionality as evidence evolves

A brilliant design no site can run delivers zero value

OBJECTIVE

Inspire differently informed clinical trial designs where optimization extends beyond power and sample size

APPROACH

Forging a deeper, mutual understanding of role and impact of statistical trial design on partnering functions through concrete, useable example: seamless Phase 2-3 design

PARTNERS IN FOCUS

Clinical Operations, Drug Supply, and IRT (Interactive Response Technology)



Workstream Outputs

Four outputs carry the workstream from research to organisational capability

01

RESEARCH

Underway

Cross-functional research into how a seamless phase 2-3 design propagates through clinical operations, drug supply, and IRT

02

MANUSCRIPT

End of 2026

Manuscript under preparation summarizing connectivity, and impact of implementation of seamless phase 2-3 design on operations, drug supply, and IRT

03

CROSS-FUNCTIONAL UPSKILLING

Organizational resource

The manuscript (and the associated research) will be used to generate focused content summaries which will improve biostatisticians' cross-functional understanding within their organizations.

04

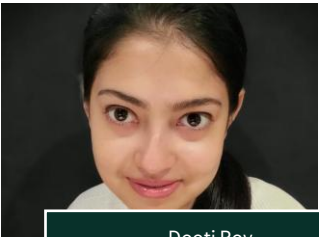
PUBLIC EDUCATION AND AWARENESS

Ongoing

Including conference presentations and webinars

Sub-Team Members

We are recruiting!



Dooti Roy
(Statistician, Boehringer Ingelheim)



Angela Zhu
(Statistician, Boehringer Ingelheim)



Kristen Daniels
(Early Asset Lead, Boehringer Ingelheim)



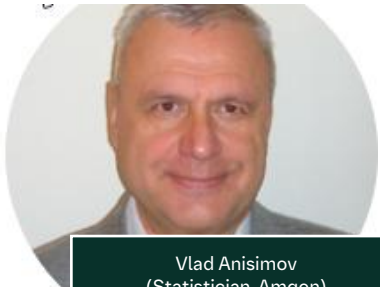
Haoyu Wang
(Statistician, BMS)



Jessica Cannon-Hill
(Statistician, Devices, J&J)



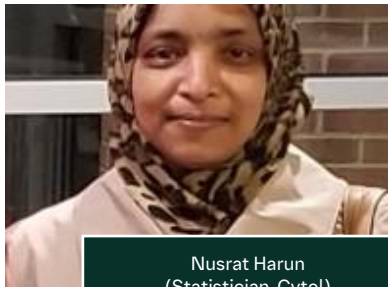
Kristen Leraas
(Clinical Operations, J&J)



Vlad Anisimov
(Statistician, Amgen)



Richard Shu
(Statistician, Genentech)



Nusrat Harun
(Statistician, Cytel)



Mary Alice Norrison
(IRT Management, Boehringer Ingelheim)

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Thank *you* for listening!